

MODELLING ANIMAL MOVEMENT FOR BEHAVIOURAL INFERENCE



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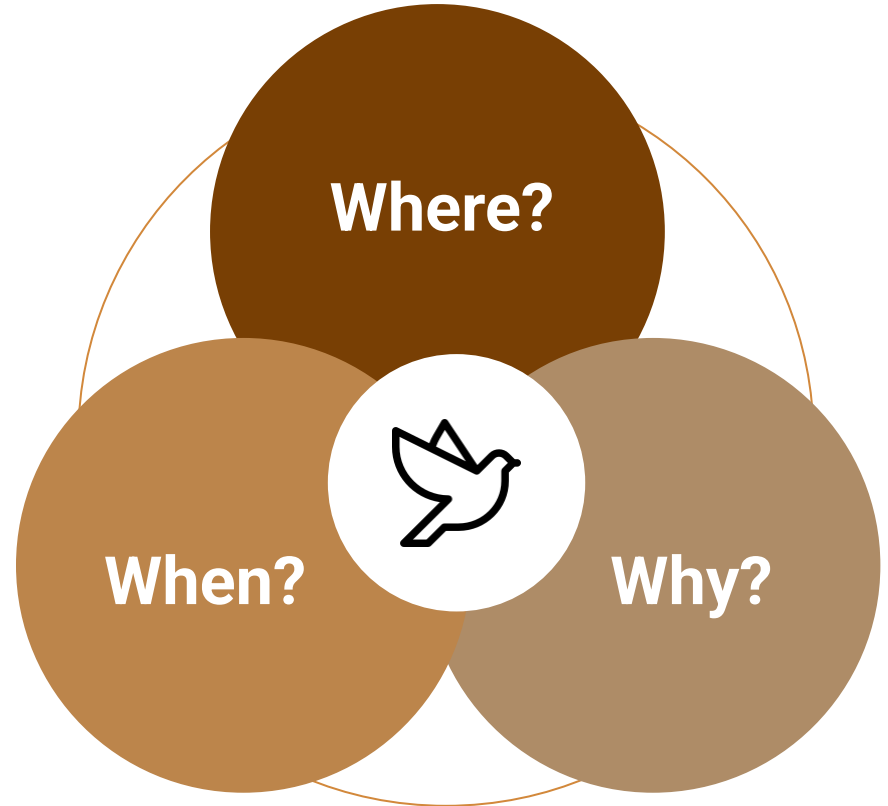
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Background

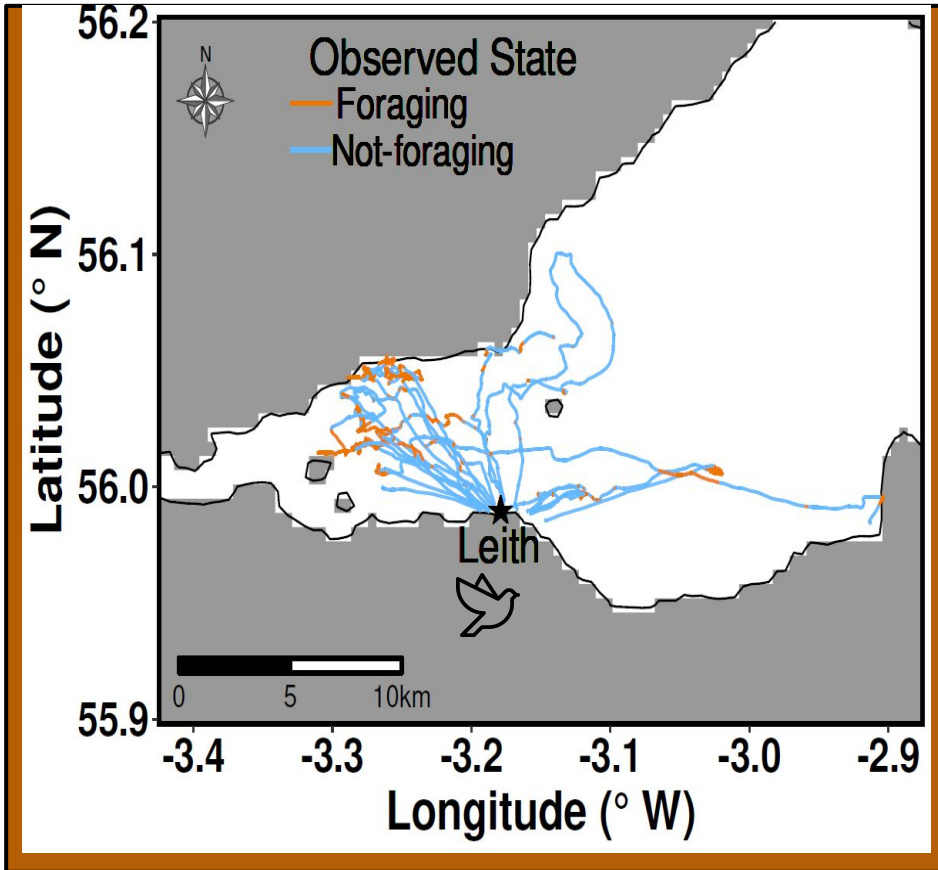


Motivation



- Seabird population decline
- The difficulty to directly observe behaviours influencing movement in the marine environment
- The need to identify at-sea seabird behaviours for conservation actions

Problem



- Observed behavioural data can be difficult to obtain
- Inferred behaviours from GPS location data are rarely validated

Goal



Validate inferred behaviours of seabirds
for effective conservation management

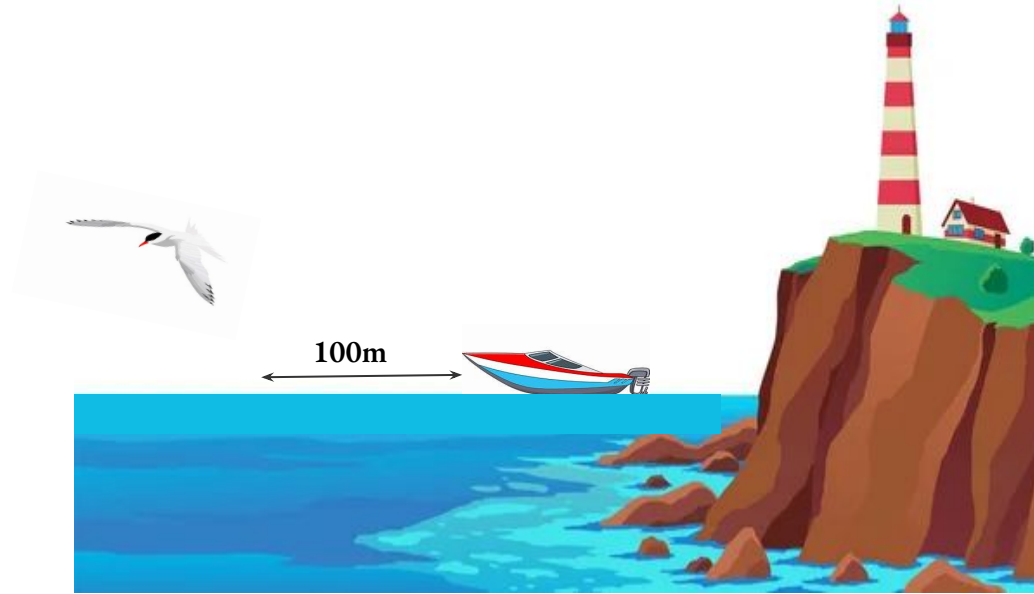
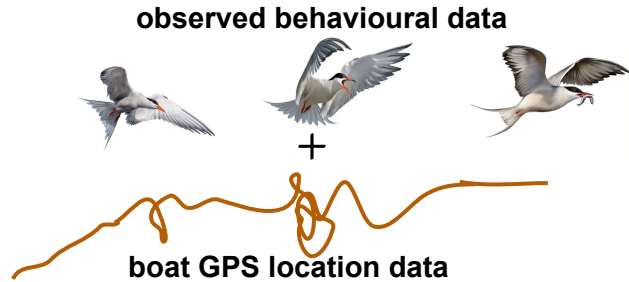
Data

Source



Method visual tracking¹

Type



Period



Year 2009-2011 Sampling interval 1-second

¹Perrow, M. R., Skeate, E. R. & Gilroy, J. J. (2011). Visual tracking from a rigid-hulled inflatable boat to determine foraging movements of breeding terns

Study site

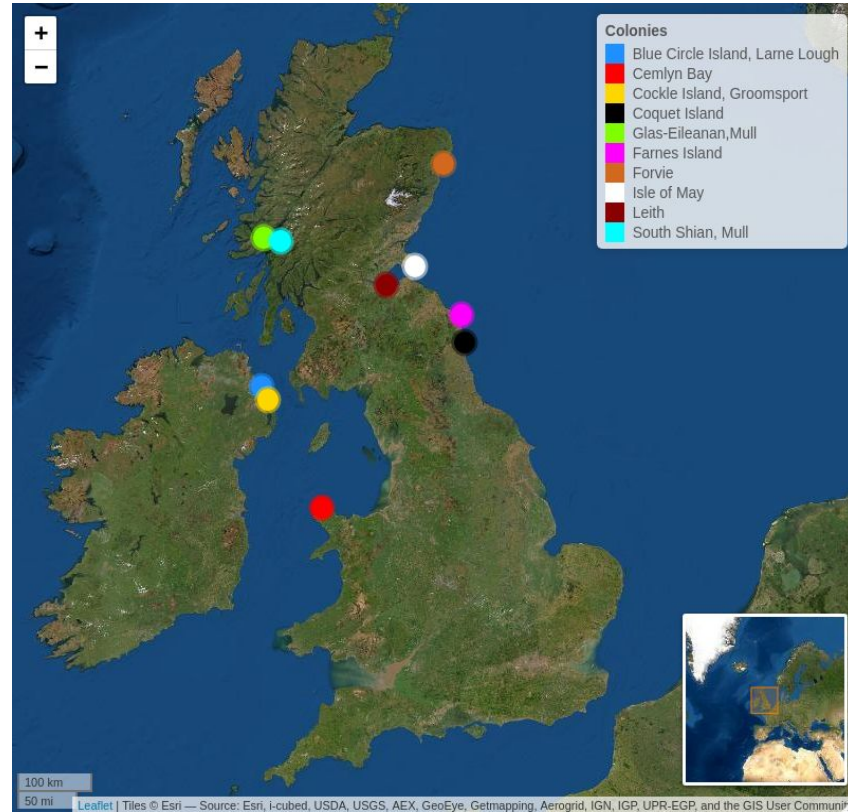


Figure: Study site consisting of 9 tern breeding colonies in the UK

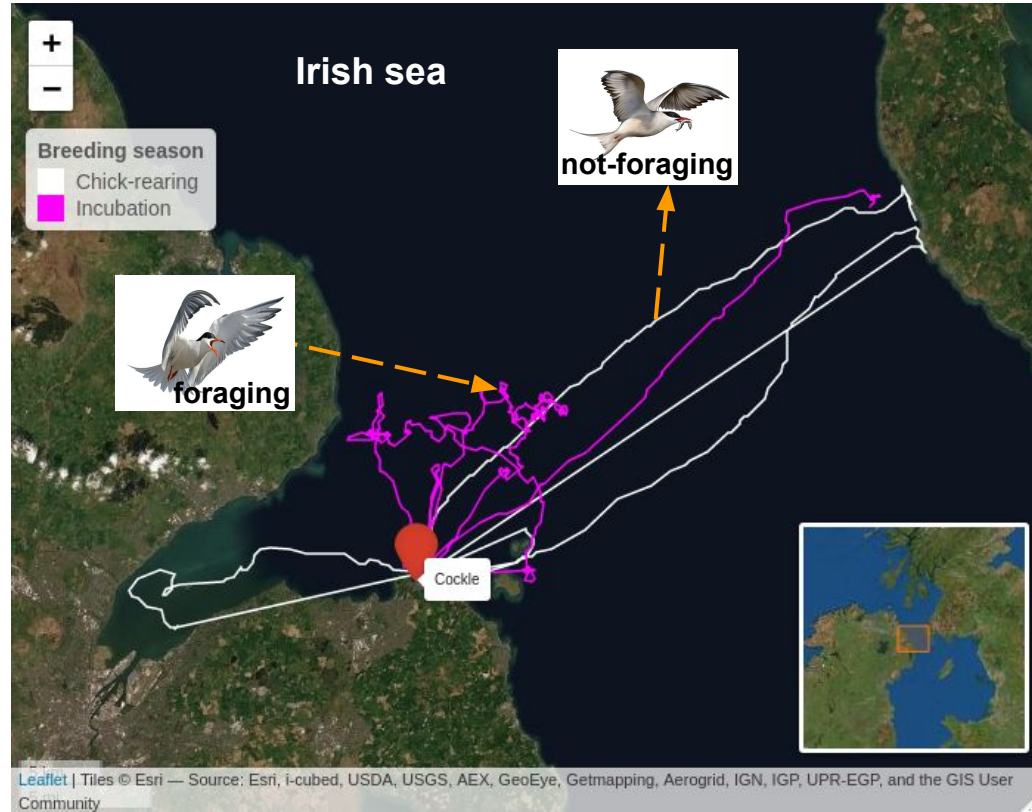


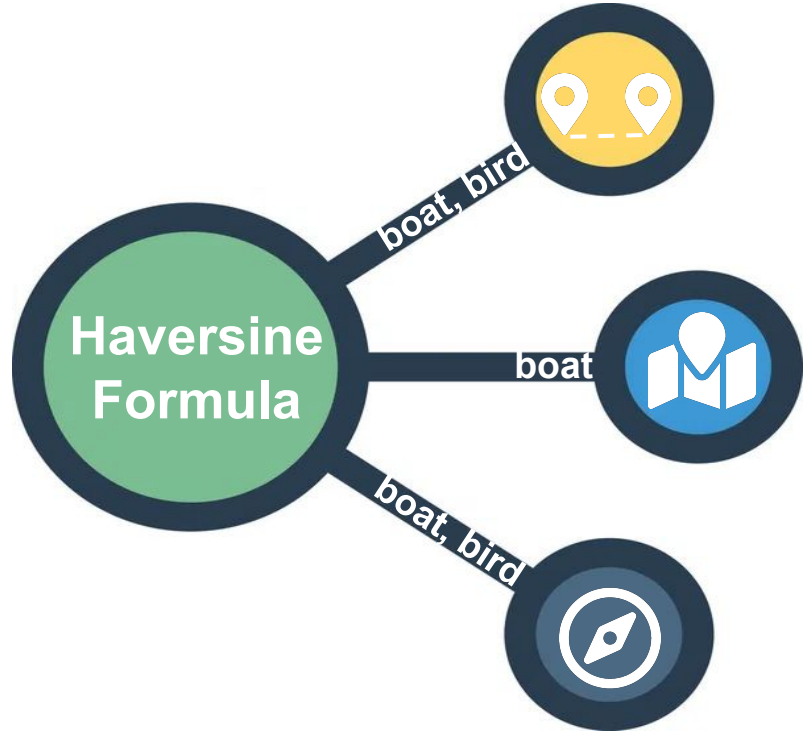
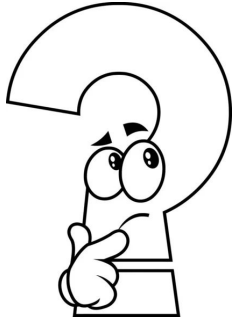
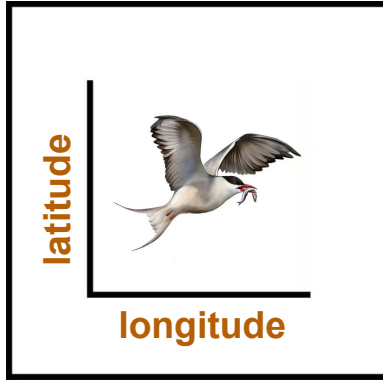
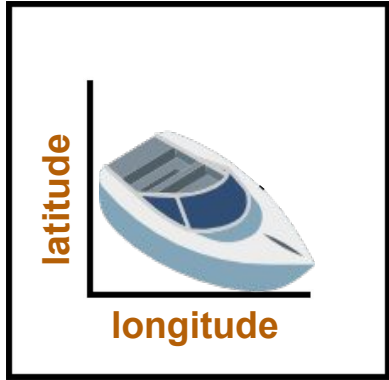
Figure: Visual tracking (boat location) data of sandwich terns at Cockle Island colony ($54^{\circ}67'N, 5^{\circ}61'W$), Northern Ireland, 2010

💡 Workflow



- 💡 **Assess** boat GPS location data as a proxy for terns location
- 💡 **Infer** behaviours from boat GPS location data using hidden Markov model
- 💡 **Validate** inferred behaviours with observed behavioural data of terns

Assessing boat location data



Results: assessing boat location data

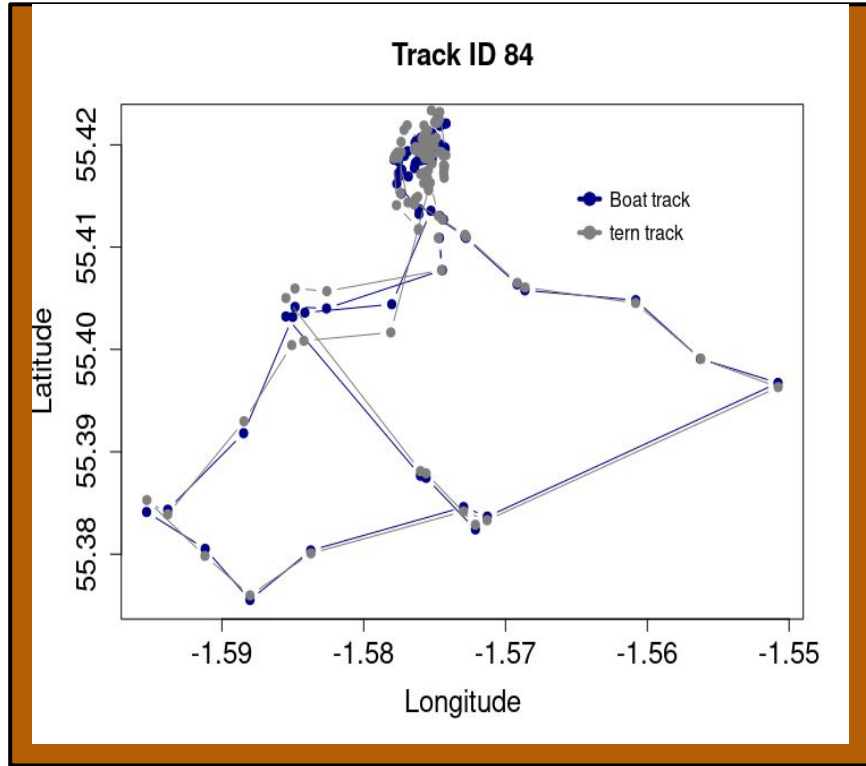


Figure: Boat and approximate tracking data of a roseate tern in Coquet Island, 2009

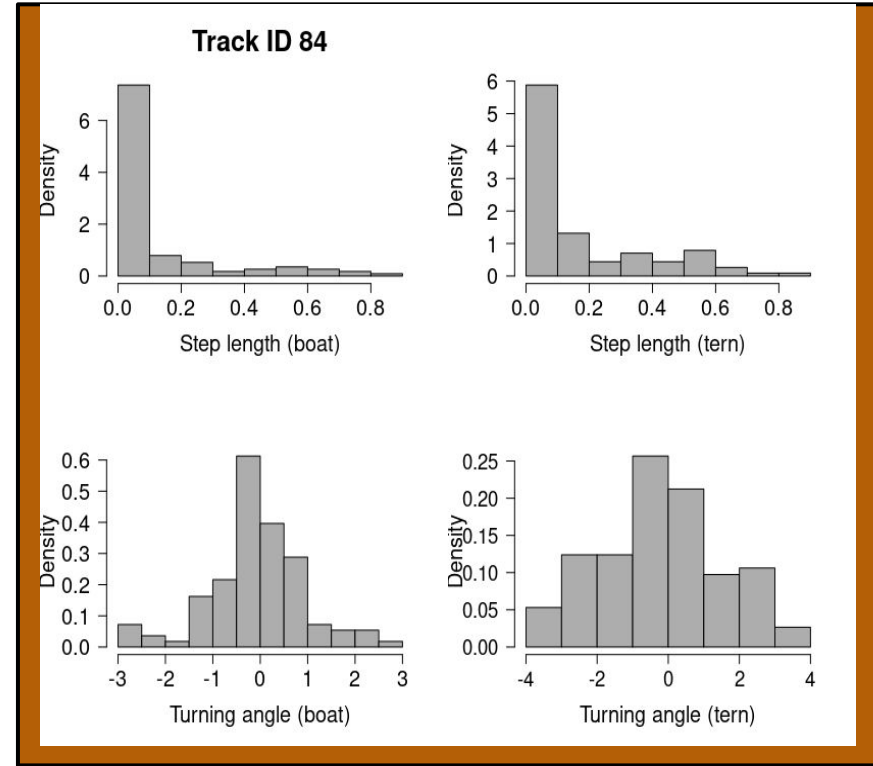


Figure: Step length (km) and turning angle (radians) distributions of a roseate tern in Coquet Island, 2009

Hidden Markov Model (HMM)

A HMM is a time series model with an **observed** component driven by an underlying **hidden** component

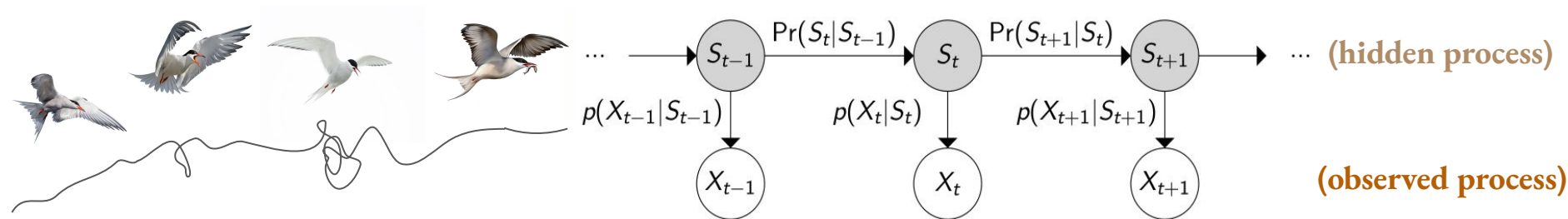


Figure: Representation of a HMM

- Foraging and Not-foraging behavioural states
- Metrics of step length and turning angles from boat GPS coordinates

Results: assessing boat location data

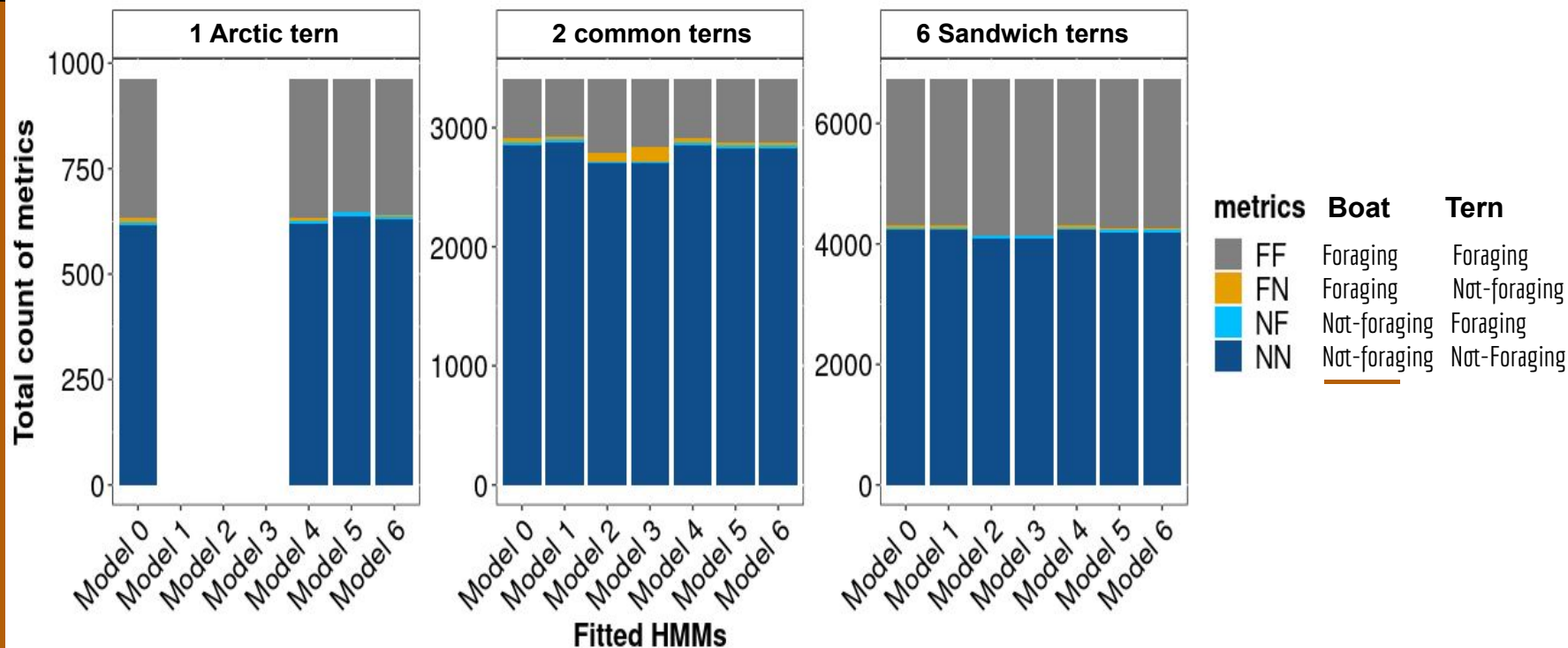


Figure: Confusion matrix metrics of behavioural states inferred from HMMs fitted to boat and tern approximate location data in Coquet colony during chick-rearing, 2009

Results: inferring behavioural states

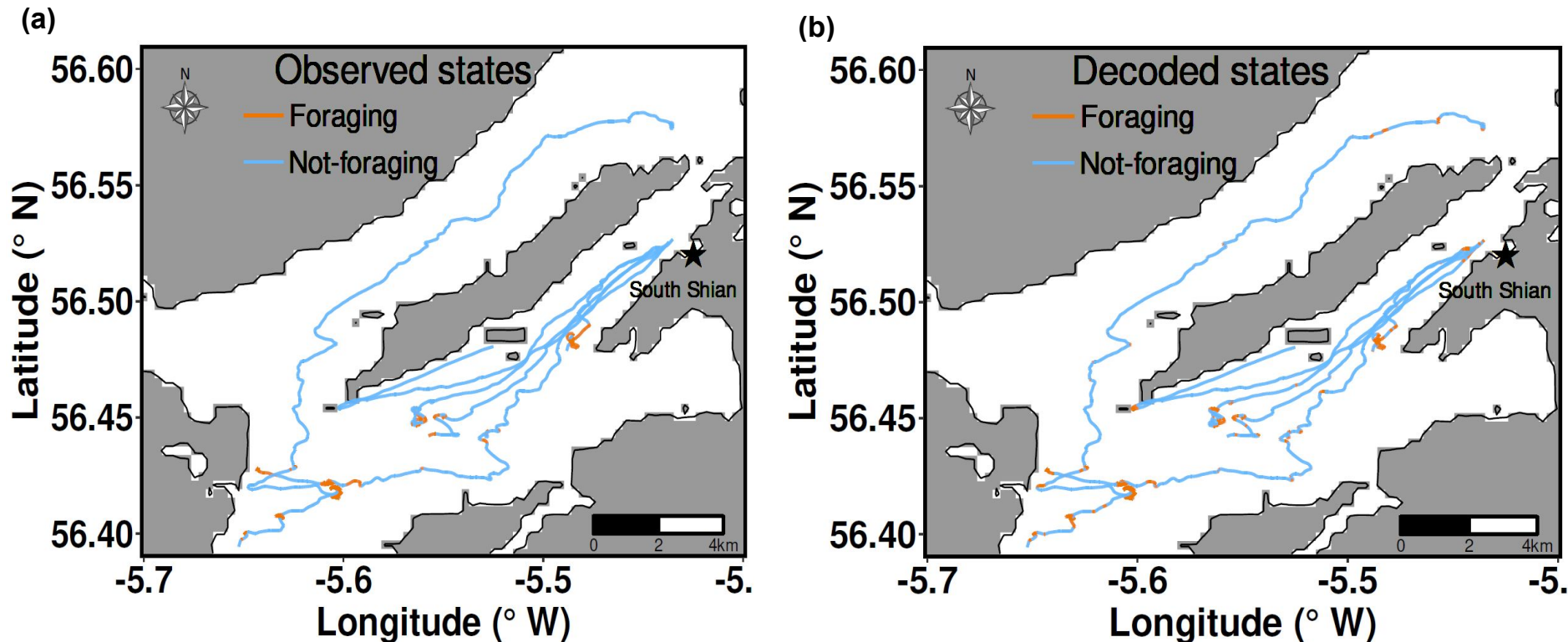


Figure: Proxy (boat) movement data of 6 visually tracked common tern coloured with (a) observed and (b) decoded behavioural states in South Shian colony, Mull, Scotland, during chick-rearing period, 2011

Results: inferring behavioural states

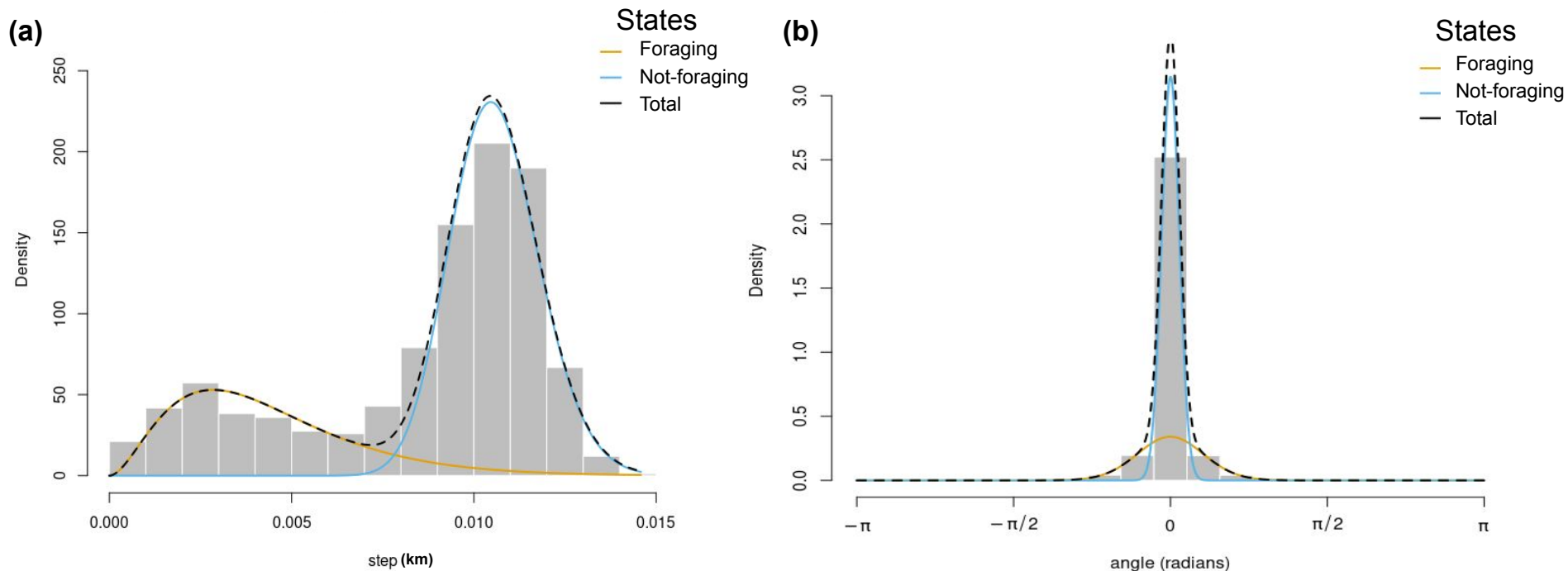


Figure: Histograms of the (a) proxy step length and (b) turning angle distributions of 6 visually-tracked common tern in South Shian colony during chick-rearing period, 2011. Lines represent the HMM fitted state-dependent distributions, and are coloured according to decoded behavioural state: **foraging** and **not-foraging**

Results: validating inferred state

- HMMs performance in decoding behavioural states
- Correctly decoded foraging behavioural state during chick rearing



Arctic tern

~~64-74%~~

58-70%

3 colonies



Common tern

~~76-84%~~

70-90%

5 colonies



Roseate tern

76%

86%

1 colony



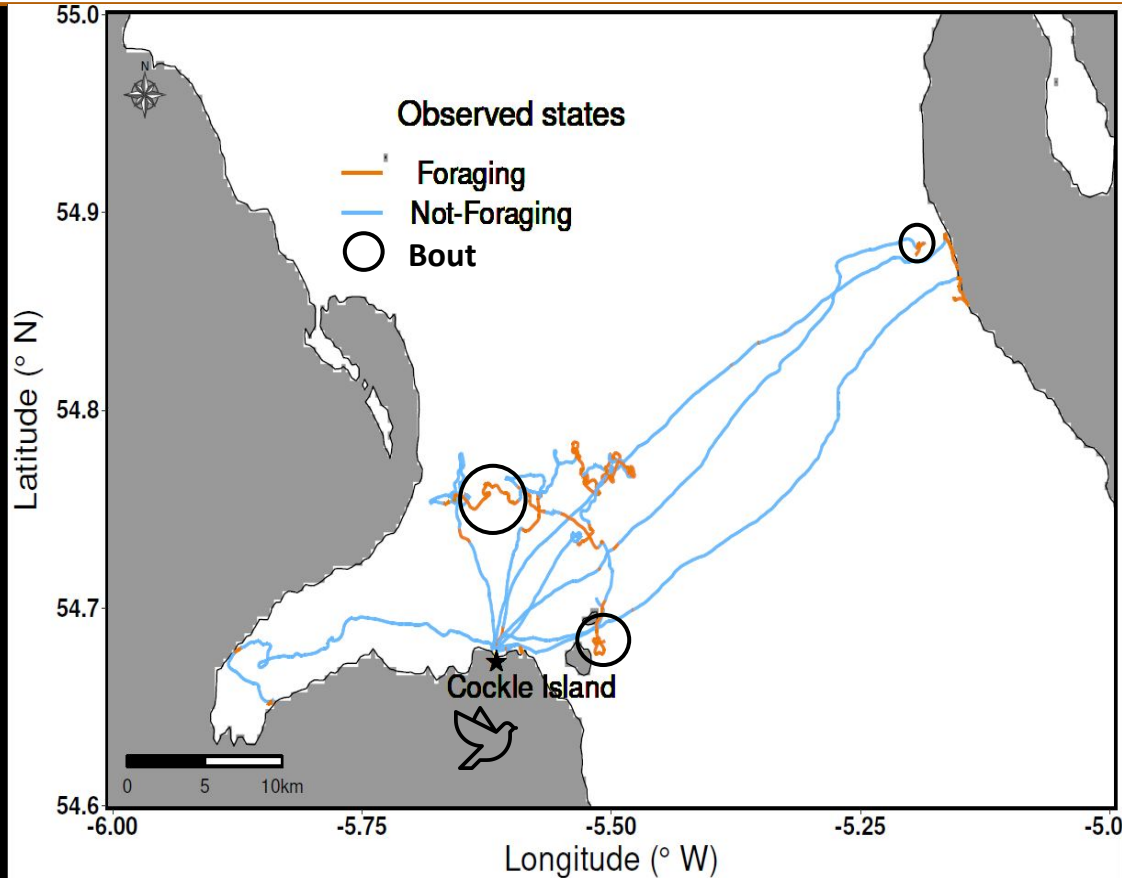
Sandwich tern

~~72-88%~~

74-91%

4 colonies

Foraging event



A **foraging event** is a bout where foraging activity occur within the tracking data

Result: foraging event in chick-rearing

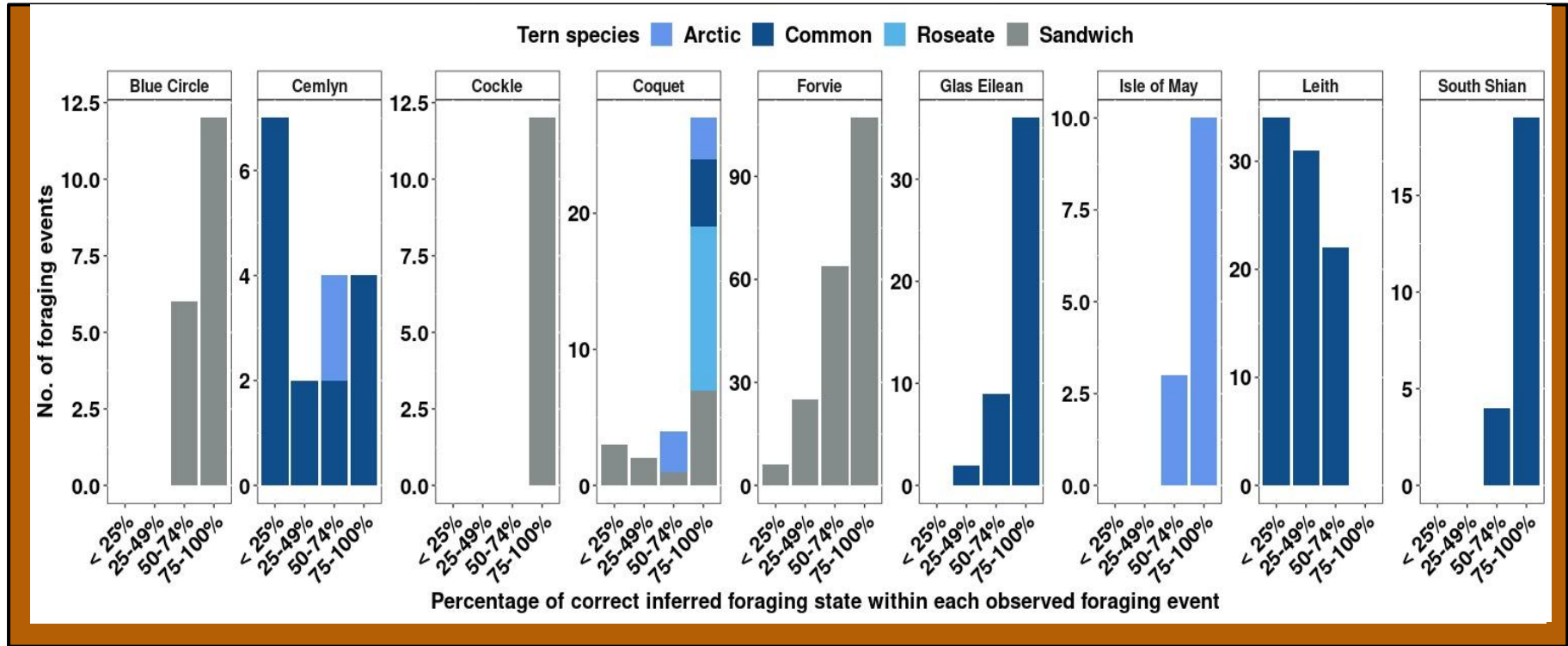


Figure: Proportion of correctly inferred foraging states captured within each observed foraging event across breeding colonies. Inferred foraging states are from HMMs deemed optimal based on validation

Result: missed foraging event

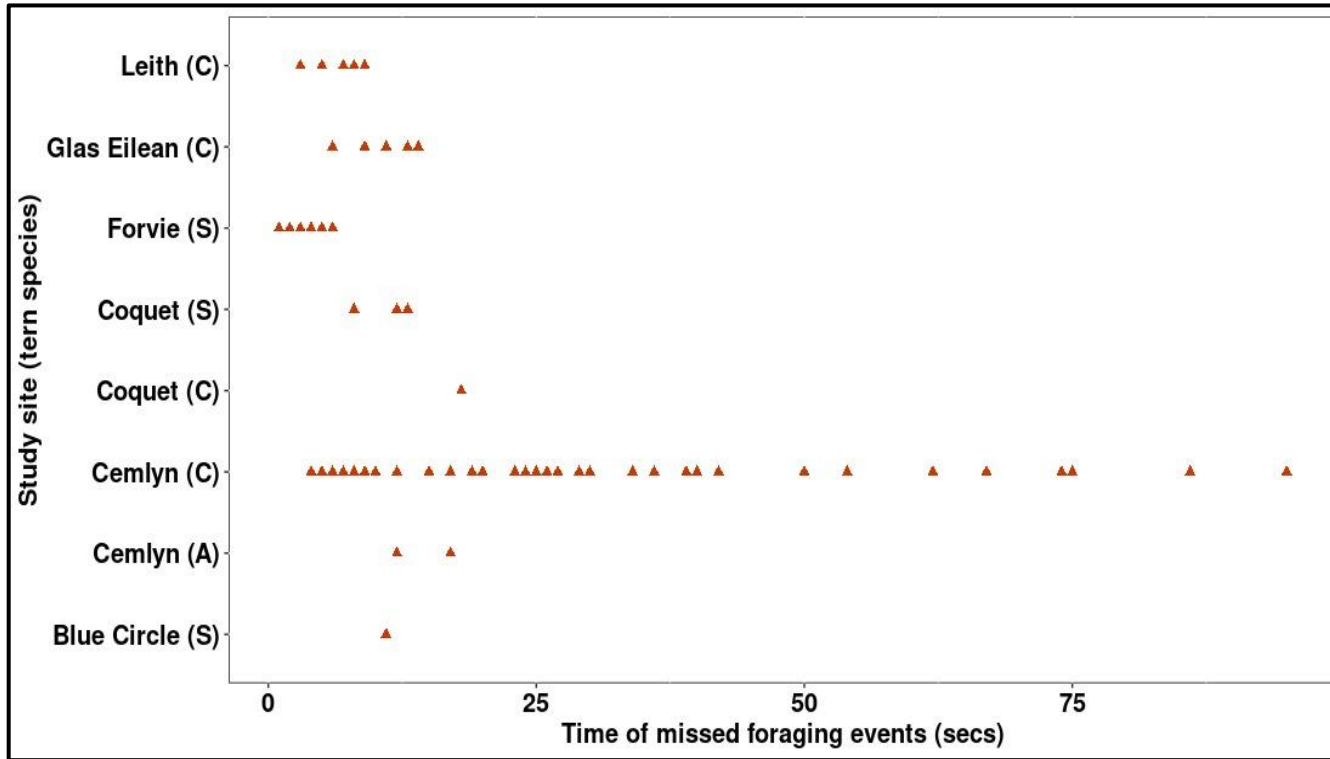


Figure: Observed foraging event completely missed from decoded behavioural states across study sites. A, C, S = Arctic, common and Sandwich tern

Summary of findings



- Visual tracking data is suitable for our study as boat tracks closely replicate those from the approximate tern tracks
- HMMs failed to capture **only few** foraging bouts within the tracking data
- Important conservation-relevant behaviours inferred using HMMs is justified

Implication of findings



- Increased confidence in unvalidated inferred behaviours from telemetry data
- Conservation-relevant behaviours inferred using HMMs can inform the planning and design of designated protected areas

Future work



Assess effect of varying temporal resolution on behavioural inference

- **Coarse temporal resolution** may miss important events such as behaviour switching
- **A finer temporal scale** comes at a cost of sampling duration and GPS battery life

References

Zucchini, W., MacDonald, I. L. & Langrock, R. (2016). **Hidden Markov Models for Time Series: An Introduction Using R**

McClintock, B. T. & Michelot, T. (2018). **momentuhmm: R package for generalized hidden Markov models of animal movement**, *Methods in Ecology and Evolution* 9 (6), 1518–1530

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Thanks for listening



Questions?